# CHITKARA UNIVERSITY INSTITUTE OF ENGINEERING & TECHNOLOGY, PUNJAB

## DEPARTMENT OF INTERDISCIPLINARY COURSES IN ENGINEERING

**Embedded Systems and Internet of Things Laboratory**

Class Group G1 | Section A | Project Group 02

**Project Abstract**

## Project Title: Smart Campus Environmental Monitoring System Using IoT

**Abstract:** This project focuses on monitoring campus environmental conditions by measuring air quality, temperature, atmospheric pressure, and rainfall using IoT technology. The system continuously gathers environmental data and processes it through an IoT-enabled microcontroller, which generates a locally accessible real-time web dashboard for instant visualization. This helps users track weather patterns, identify polluted areas, and observe changing atmospheric conditions across the campus. The solution is cost-effective, easy to deploy, and suitable for institutions aiming to promote environmental awareness and sustainability.

**Application Area:**

- Campus Environmental Monitoring
- Pollution Analysis and Control
- Smart and Sustainable Campuses

**Impact on Society/Industry:**

- Healthier Living and Working Environment: Continuous monitoring helps maintain better air quality.
- Reduction of Carbon Footprint: Helps reduce carbon footprint through data-based monitoring.
- Support for Sustainable Infrastructure: Supports sustainable and smart campus or industrial environments.

**Sustainable Development Goals:** SDG 3 – Good Health and Well-Being, SDG 9 – Industry, Innovation and Infrastructure, SDG 11 – Sustainable Cities and Communities, SDG 13 – Climate Action.

**Technology Stack:** ESP32, IoT, Environmental Sensors, Embedded C++, Cloud/IoT Platform

**Group Details:**

| Sr. No | Name of Students | Roll No | Project Guide Name and Signature |
|---|---|---|---|
| 1. | Abhay Singh Pundir | 2410990006 | Dr. Manvinder Sharma (ES&IOT Lab Faculty) |
| 2. | Abhinav Singh | 2410990009 | |
| 3. | Abhinav Verma | 2410990010 | |
| 4. | Abhiraj Dhiman | 2410990011 | |
| 5. | Abhishek Sharma | 2410990012 | |

**Approved By:**

| Name of Faculty in Panel | Designation | Signature |
|---|---|---|
| (Lab Internal Faculty Name) | Designation of Lab Internal Faculty | |
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**Final Approved by**
**Dr. Rajneesh Talwar**
**Dean DICE**

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## DEPARTMENT OF INTERDISCIPLINARY COURSES IN ENGINEERING

**Embedded Systems and Internet of Things Laboratory**

Class Group G1 | Section A | Project Group 02

**Project Abstract**

## Project Title: Campus Air quality, Carbon footprint, temperature monitoring using IOT

**Abstract:** This project focuses on monitoring campus environmental conditions by measuring air quality, carbon footprint, and temperature using IoT technology. Various sensors are deployed to continuously collect data related to air pollutants, carbon emission levels, and ambient temperature across campus. The collected data is processed using a microcontroller and transmitted to a cloud or monitoring platform for real-time visualization and analysis. This system helps in identifying pollution-prone areas and understanding the impact of human activities on the campus environment. The proposed solution is cost-effective, easy to implement, and suitable for educational institutions aiming to promote environmental awareness and sustainability.

**Application Area:**

- Campus Environmental Monitoring
- Pollution Analysis and Control
- Smart and Sustainable Campuses

**Impact on Society/Industry:**

- Healthier Living and Working Environment: Continuous monitoring helps maintain better air quality.
- Reduction of Carbon Footprint: Helps reduce carbon footprint through data-based monitoring.
- Support for Sustainable Infrastructure: Supports sustainable and smart campus or industrial environments.

**Sustainable Development Goals:** SDG 3 – Good Health and Well-Being, SDG 9 – Industry, Innovation and Infrastructure, SDG 11 – Sustainable Cities and Communities, SDG 13 – Climate Action.

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